Hadoop & Spark - ISEN 2025/2026

Lab: Dataframes

For this Lab you will need:

- The 3.4 Lab

As a reminder

Check 3.1 to login in NGROK, Kaggle, install Java or Spark and import libraries.

1 Create a Dataframe using an existing RDD

Lets create a simple Dataframe from an Rdd

```
# RDD:
rdd = sc.parallelize(
    [
        [1,"one"],
        [2,"two"],
        [3,"three"],
        [4,"four"],
        [5,"five"],
     ]
    )
# Dataframe:
df = rdd.toDF()

df.show()
```

Manipulate Dataframes

Notice that columns have a no name, you need to specify them

```
dfFixed = rdd.toDF(["Id","Name"])
```

Lets show it:

```
dfFixed.show()
```

Manipulate Dataframes

You can see the schema like this. Notice the types were infered correctly automatically:

```
dfFixed.printSchema()
```

2 Create a Dataframe reading a file

Lets read from a file.

```
fileDF = spark.read.csv('lego_sets_and_themes.csv') # Note: not the cleaned ONE
```

Note that this operation is very quick: as a reminder, reading a file is a transfromation, so it will only be done when an action is done, for example a show:

```
fileDF.show()
```

Manipulate Dataframes

You can specify the columns manually or declare that the header is in the first row:

```
# Note: not the cleaned one but the original
fileDF = spark.read.option("header","true").csv('lego_sets_and_themes.csv')
fileDF.printSchema()
```

One issue here, the types are not infered correctly (all strings).

We will need to force the schema.

Schema

When reading we can ask spark to infer the schema, but this does not always work as you can see/

```
# Note: not the cleaned one but the original
fileDF = spark.read.option("header","true").option("inferShema","true").csv('lego_sets_and_themes.csv')
fileDF.printSchema()
```

Manipulate Dataframes

In a production environment, it is better to declare the schema manually for 2 reasons:

- Scanning data to infer the data every time can slow down your pipeline
- But more importantly, it can create errors when types are not recognized or are changing

Manipulate Dataframes

Lets declare manually this schema using a StructType object and read our cleaned file:

```
from pyspark.sql.types import *

manualSchema = StructType(
    [
        StructField("set_number",      StringType(),  True),
        StructField("set_name",        StringType(),  True),
        StructField("year_released",   IntegerType(), True),
        StructField("number_of_parts", IntegerType(), True),
        StructField("image_url",       StringType(),  True),
        StructField("theme_name",      StringType(),  True)
    ]
)
```

Manipulate Dataframes

We wiil use the cleaned file from previous lab, if you do not have it anymore you can use:

```
df = spark.read.csv('lego_sets_and_themes.csv', header=True, sep=",")
df = df.withColumn("number_of_parts", df["number_of_parts"].cast(IntegerType()))
df = df.withColumn("year_released", df["year_released"].cast(IntegerType()))
df.write.options(delimiter=';').mode("overwrite").csv("lego_sets_and_themes_clean.csv")
```

Manipulate Dataframes

Apply the Schema

```
schemaDF = spark.read
  .option("header","true")
  .option("delimiter",";")
  .schema(manualSchema)
  .csv('lego_sets_and_themes_clean.csv')

schemaDF.printSchema()
```

And Show:

```
schemaDF.show(5)
```

Exercice

In this exercise, you will use the "Video Game Sales" dataset from Kaggle to practice working with PySpark DataFrames. The dataset contains information about video game sales, such as the platform, genre, publisher, and global sales figures.

You can download the dataset here: https://www.kaggle.com/gregorut/videogamesales

Follow the steps below to complete the exercise:

- 1 Download the dataset and upload it to your working environment.
- 2 Define the schema for the dataset
- 3 Read the dataset using the defined schema
- 4 Display the first 10 rows of the DataFrame
- 5 Print the schema of the DataFrame
- Bonus: Find the top 10 best-selling games of all time

> **Hint:** for the bonus, use the command `select` and `orderBy`